



# V93000 EXA Scale EX Specifications

## EXA Scale EX Infrastructure

**Version 8.6.0**

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# Notices

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# Infrastructure specifications - V93000 EXA Scale EX systems

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## *Scope of specifications*

Advantest specifies and verifies the specifications at the DUT interface pogo level. Using an adequate DUT board, valid system calibration and a fixture delay measurement (TDR), these specifications are valid at the device under test as well. Specifications describe warranted product performance. Characteristics are included as typical values to provide additional useful information by describing typical non-warranted performance.

## *Test programming and debug*

The SmartTest production test environment includes manufacturing test interfaces to operators, handling equipment and factory networks. A graphical testflow editor links device tests into a production-ready test program, where the tests are set up via fill-in-the-blank test functions. User-specific tests are programmed with test methods in Java. Links are available for design-to-test conversion. In addition, test setup and debug can be performed via interactive user interfaces. Result analysis tools are available for error locations, timing behavior, analog waveforms, shmoo plots, pin margin tests and memory bit map.

## *Supported channel cards*

The following native V93000 EXA Scale channel cards are supported:

- XPS256
- PS5000
- Link Scale USB3

## *EXA Scale - Xtreme Link*

The V93000 EXA Scale system employs Xtreme Link technology, a communication network designed specifically for Automatic Test Equipment (ATE). The technology provides high-speed data connections, embedded computing power, and instant card-to-card communication.

- **Specialized ATE network:** Xtreme Link technology enables the highest bandwidth for large test data which leads to the fast data transfer across ATE pins and to workstation, reduction test time and Instant pin to pin networking. Instant communication between all pins enables fast response to device conditions, alarms, Instrument to instrument synchronization and real time site shutdown.
- **Digital upload performance:** EXA Scale Xtreme Link dramatically improves the digital capture result upload performance.
- **Scan diagnosis:** EXA Scale Xtreme Link significantly improves uploads scan results.
- **Xtreme Pooling:** Flexible memory pooling and sharing across entire card without application bandwidth limitation for efficiently using on-board memory independent of pin usage enabled by Xtreme Link.
- **Unlimited ganging:** EXA Scale Xtreme Link enables unlimited ganging of DPS channels across cards without compromising accuracy.

## *Compatibility*

Software is compatible between all test heads.

The same DUT boards fit on all test heads. The docking system is identical for all test heads. The detailed docking pin positions might differ between different test heads. For details check the documentation on docking.

**Operating system**

Linux Red Hat Enterprise is used in all test head configurations.

**Calibration and operation**

|                                                 |                         |
|-------------------------------------------------|-------------------------|
| Warm-up time                                    | 60 minutes              |
| Basic maintenance period                        | 6 months                |
| Base calibration period (traceable calibration) | 6 months                |
| Calibration period (system auto adjustment)     | 3 months <sup>[1]</sup> |

**Further information**

For more information about the Advantest V93000 SOC Series, please visit [www.advantest.com](http://www.advantest.com).

**Contact information**

For more information about the Advantest V93000 SOC cards, please contact your local Advantest sales representative.

This information is subject to change without notice.

<sup>[1]</sup> Valid at ambient temperature within  $\pm 2.5$  K of calibration temperature.

## Test head

| Maximum card count   | EXA Scale EX test head |
|----------------------|------------------------|
| Number of card cages | 1                      |
| Number of slots      | 3                      |
| Supported cards      | Maximum channel count  |
| XPS256               | 768                    |
| PS5000               | 3x256                  |

## System slots

Supported slots

| Maximum card count   | EXA Scale EX test head |
|----------------------|------------------------|
| Number of card cages | 1                      |
| Max. number of slots | 3                      |

<sup>[2]</sup> No Utility Card installed, using Internal Utility Line option.

## EXA Scale EX device power supplies

Supported channel slot device power supplies

| Maximum channel count | EXA Scale EX test head |
|-----------------------|------------------------|
| Number of slots       | 3                      |
| DC Scale XPS256       | 768                    |

Note

Mixing different device power supply types within one Test Head is supported.  
Refer to DC Scale device power supply specification sheet for more information.



## Parametric measurements

### Direct DVM Access (at utility pogo block)

A DVM can be routed through the DC rail onto each tester channel.

#### Supported Digital Multimeter (DMM)

Keysight 3458A with option 002

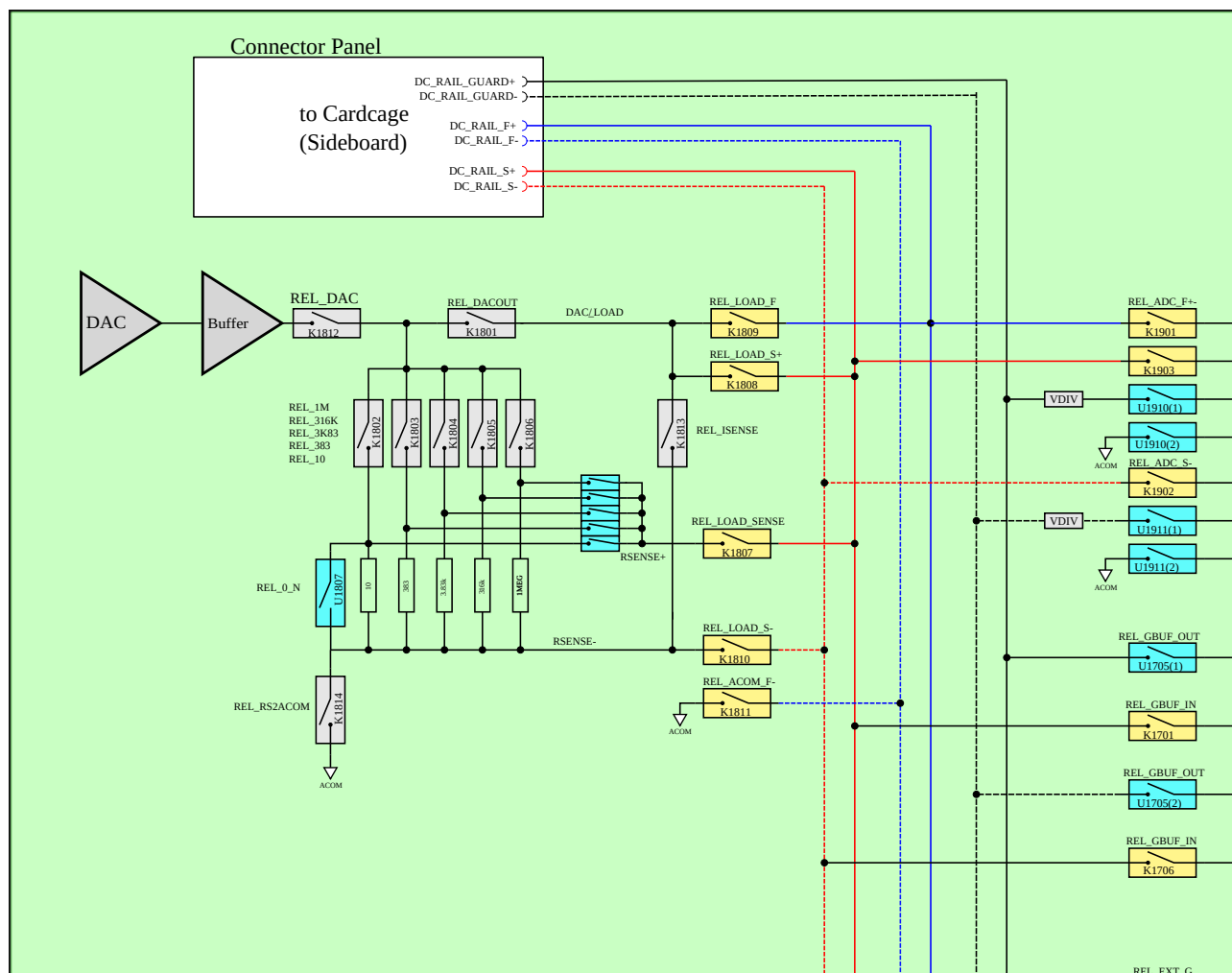
#### Absolute voltage measurement accuracy

$\pm 40\mu\text{V}$  + DMM specification

#### Input voltage range

-5 V to 7.5 V

The diagram of the system control board is shown below.



The system control board diagram

## Utility resources

Utility resources are the utility power supplies and the utility lines that you can use to supply and control additional hardware on the DUT board.

Utility lines of V93000 EXA Scale systems allow you to control external devices such as relays, solenoids or indicators, depending on the application. The lines are implemented as FET switch outputs.

The utility power supplies of V93000 EXA Scale systems support DUT boards with additional active components. Typical examples for these active components are relays, switches, operational amplifiers, and buffers.

### Internal utility resources

Internal utility resources of a V93000 EXA Scale system in the basic configuration consist of 64 statically controlled utility lines, a +5 V utility power supply and the Internal Utility Power Option.

#### Internal utility lines

In the following table, only the "Basic configuration" columns with "Optional Utility Card (EK051A)" set to "none" apply to the internal utility lines. The other columns require the installation of minimum one optional *Utility Card (EK051 and EK051A)* on page 12 that replaces the internal utility lines.

| Configuration: Basic configuration              | Utility Card configuration |
|-------------------------------------------------|----------------------------|
| TH: EXA Scale EX                                | EXA Scale EX               |
| Slots: 3                                        | 2 <sup>[3]</sup>           |
| Internal utility modules: 2                     | not usable                 |
| Optional Utility Cards (EK051 and none EK051A): | 1                          |

#### Configuration

|                         |                         |                               |
|-------------------------|-------------------------|-------------------------------|
| Number of utility lines | 64                      | 256                           |
| Pogo block              | UTIL 1, 3<br>(165, 167) | UTIL 1 ... 8<br>(165 ... 172) |

Note: The 4-row utility pogo blocks of the second Utility Card replace all existing 3-row DPS-slot pogo blocks.

The DPS block sequence of the second Utility Card differs from the UTIL block sequence of the first card. For the correct pogo block assignment of the second card, refer to "Utility pogo block mapping" in TDC topic "Utility pogo block - Utility Card (V93000 EXA Scale systems)" (347642).

<sup>[3]</sup> The test head has 3 slots but one slot is occupied by the utility card.

| Configuration: Basic configuration              |    | Utility Card configuration |
|-------------------------------------------------|----|----------------------------|
| TH: EXA Scale EX                                |    | EXA Scale EX               |
| Slots: 3                                        |    | 2 <sup>[3]</sup>           |
| Internal utility modules: 2                     |    | not usable                 |
| Optional Utility Cards (EK051 and EK051A): none |    | 1                          |
| Sequencer control                               | no | yes                        |

**Output**

|                                        |                                                                                                                                                                                                     |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Type                                   | Open drain                                                                                                                                                                                          |
| Maximum current                        | 100 mA per line                                                                                                                                                                                     |
| Cut-off line current                   | 130 ±15 mA after 250 µs                                                                                                                                                                             |
| Voltage                                |                                                                                                                                                                                                     |
| Output switch open,<br>no load applied | Each utility line has an internal pull-up resistor of 10 kΩ to the internal supply voltage of 3.3 V.<br>We recommend using external pull-up resistors for applications with different requirements. |
| Output switch closed                   |                                                                                                                                                                                                     |
| $V_{ds}$                               | 0.5 V maximum                                                                                                                                                                                       |
| $R_{on}$                               | < 5 Ω                                                                                                                                                                                               |

**Input**

|                         |                                    |          |
|-------------------------|------------------------------------|----------|
| Type                    | TTL                                |          |
| Voltage range           | 0 V ... 5.0 V                      |          |
| Maximum voltage rating  | 5.5 V, internal clamp diode to 5 V |          |
| Maximum clamp current   |                                    |          |
| per channel             | 200 mA                             | 200 mA   |
| total                   | 800 mA                             | 800 mA   |
|                         | per utility module                 | per card |
| Input impedance         | 10 kΩ typ.                         |          |
| Input thresholds        |                                    |          |
| Maximum V <sub>IL</sub> | 1.3 V typ.                         |          |
| Minimum V <sub>IH</sub> | 2.1 V typ.                         |          |

**Internal utility power supplies****Utility power supply - UP5V pins**

In the following table, only the "Basic configuration" columns with "Optional Utility Cards (EK051A)" set to "none" apply to the internal UP5V utility power supply. The other columns require the installation of minimum one optional *Utility Card (EK051 and EK051A)* on page 12 that replaces the internal UP5V utility power supply.

<sup>[3]</sup> The test head has 3 slots but one slot is occupied by the utility card.

|                                                        |                                   |
|--------------------------------------------------------|-----------------------------------|
| <b>Configuration: Basic configuration</b>              | <b>Utility Card configuration</b> |
| <b>TH: EXA Scale EX</b>                                | <b>EXA Scale EX</b>               |
| <b>Slots: 3</b>                                        | <b>2 <sup>[3]</sup></b>           |
| <b>Internal utility modules: 2</b>                     | <b>none</b>                       |
| <b>Optional Utility Cards (EK051 and EK051A): none</b> | <b>1</b>                          |

**Configuration**

|                             |                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------------|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Total number of supplies    | 1                       | See <i>utility card</i> specification                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Supply routed to pogo block | UTIL 1, 3<br>(165, 167) | UTIL 1 ... 8<br>(165 ... 172)<br><br>Note: The 4-row utility pogo blocks of the second Utility Card replace all existing 3-row DPS-slot pogo blocks.<br><br>The DPS block sequence of the second Utility Card differs from the UTIL block sequence of the first card. For the correct pogo block assignment of the second card, refer to "Utility pogo block mapping" in TDC topic "Utility pogo block - Utility Card (V93000 EXA Scale systems)" (347642). |

**Characteristics per supply**

|                          |                                                                                                                                            |                                       |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| Output voltage           | +5.0 V $\pm$ 2 % sensed to the pogo pin                                                                                                    | See <i>utility card</i> specification |
| Total output current     | 4.5 A                                                                                                                                      | See <i>utility card</i> specification |
| Maximum load capacitance | 10000 $\mu$ F                                                                                                                              | See <i>utility card</i> specification |
| Pogo pins                | +5.0 V: UP5V<br>GND: UGND<br><br>Note: UP5V pins of different card cages or Utility Cards must not be connected together on the DUT board. | See <i>utility card</i> specification |

**Utility Card (EK051 and EK051A)**

The Utility Card provides 256 utility lines and several utility power supplies on eight 4-row pogo blocks. The cards can reside in any slot of a V93000 EXA Scale system. There are two variants available, EK051 and EK051A. The EK051 Utility card is used in SmarTest 8 only. The EK051A Utility card is used in both SmarTest 7 (SmarTest version 7.10 and higher) and SmarTest 8.

Here we describe utility card specification when one utility card is installed in EXA Scale EX test head. In this configuration, the internal utility modules are not useable and since one slot is occupied by the utility card the other two slots will be available.

### Utility lines

| Utility lines                          |                                                                                                                                                                                                     |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Configuration</b>                   |                                                                                                                                                                                                     |
| Number of utility lines                | 256                                                                                                                                                                                                 |
| Pogo block                             | UTIL 1 ... 8<br>(165 ... 172)                                                                                                                                                                       |
| Sequencer control                      | yes                                                                                                                                                                                                 |
| <b>Output</b>                          |                                                                                                                                                                                                     |
| Type                                   | Open drain                                                                                                                                                                                          |
| Maximum current                        | 100 mA per line                                                                                                                                                                                     |
| Cut-off line current                   | 130 ±15 mA after 250 µs                                                                                                                                                                             |
| Voltage                                |                                                                                                                                                                                                     |
| Output switch open,<br>no load applied | Each utility line has an internal pull-up resistor of 10 kΩ to the internal supply voltage of 3.3 V.<br>We recommend using external pull-up resistors for applications with different requirements. |
| Output switch closed                   |                                                                                                                                                                                                     |
| $V_{ds}$                               | 0.5 V maximum                                                                                                                                                                                       |
| $R_{on}$                               | < 5 Ω                                                                                                                                                                                               |
| <b>Input</b>                           |                                                                                                                                                                                                     |
| Type                                   | TTL                                                                                                                                                                                                 |
| Voltage range                          | 0 V ... 5.0 V                                                                                                                                                                                       |
| Maximum voltage rating                 | 5.5 V, internal clamp diode to 5 V                                                                                                                                                                  |
| Maximum clamp current                  |                                                                                                                                                                                                     |
| per channel                            | 200 mA                                                                                                                                                                                              |
| total                                  | 800 mA                                                                                                                                                                                              |
| Input impedance                        | 10 kΩ typ.                                                                                                                                                                                          |
| Input thresholds                       |                                                                                                                                                                                                     |
| Maximum $V_{IL}$                       | 1.3 V typ.                                                                                                                                                                                          |
| Minimum $V_{IH}$                       | 2.1 V typ.                                                                                                                                                                                          |

### Utility power supplies

#### Utility power supply - UP5V pins

| UP5V pins                |   |
|--------------------------|---|
| <b>Configuration</b>     |   |
| Total number of supplies | 1 |

| UP5V pins                                                                                                 |                                         |
|-----------------------------------------------------------------------------------------------------------|-----------------------------------------|
| Supply routed to pogo block                                                                               | UTIL 1 ... 8<br>(165 ... 172)           |
| <b>Characteristics per supply</b>                                                                         |                                         |
| Output voltage                                                                                            | +5.0 V $\pm$ 2 % sensed to the pogo pin |
| Total output current                                                                                      | 8.0 A                                   |
| Maximum load capacitance                                                                                  | 10000 $\mu$ F                           |
| Pogo pins                                                                                                 | +5.0 V: UP5V                            |
|                                                                                                           | GND: UGND                               |
| Note: UP5V pins of different card cages or Utility Cards must not be connected together on the DUT board. |                                         |

### Utility power supply - XP pins

#### Configuration

| XP pins                     |                               |
|-----------------------------|-------------------------------|
| Total number of supplies    | 3                             |
| Supply routed to pogo block | UTIL 1 ... 8<br>(165 ... 172) |

#### Characteristics per power supply

| Voltage        | Accuracy  | Maximum current | Max. load capacitance | Label  |
|----------------|-----------|-----------------|-----------------------|--------|
| +5.0 V, sensed | $\pm$ 2 % | 12.8 A          | 10000 $\mu$ F         | XPP5V  |
| -5.0 V, sensed | $\pm$ 2 % | 6 A             | 7000 $\mu$ F          | XPM5V  |
| +3.3 V, sensed | $\pm$ 2 % | 8 A             | 10000 $\mu$ F         | XPP3V3 |

### Utility power supply - UPP15V / UPM15V / F24V pins

#### Configuration

| UPP15V / UPM15V / F24V pins  |        |                        |
|------------------------------|--------|------------------------|
| Total number of supplies     |        | 4                      |
| Supply routed to pogo block  |        |                        |
| Voltage                      | Label  |                        |
| +15 V                        | UPP15V | UTIL 2 + 8 (166 + 172) |
| -15 V                        | UPM15V | UTIL 3 + 5 (167 + 169) |
| 24 V floating <sup>[4]</sup> | F24VA  | UTIL 4 (168)           |
|                              | F24VB  | UTIL 6 (170)           |

<sup>[4]</sup> Floating range:  $\pm$ 24 V max.

## Characteristics per power supply

| Voltage                                  | Accuracy | Maximum current | Max. load capacitance | Label           |
|------------------------------------------|----------|-----------------|-----------------------|-----------------|
| +15.0 V, sensed                          | ±2 %     | 3 A             | 1000 µF               | UPP15V          |
| -15.0 V, sensed                          | ±2 %     | 3 A             | 1000 µF               | UPM15V          |
| +24.0 V, sensed, floating <sup>[4]</sup> | ±4 %     | 2 * 1 A         | 200 µF                | F24VA and F24VB |

**Note** The 4-row utility pogo blocks of the second Utility Card replace all existing 3-row DPS-slot pogo blocks.

The DPS block sequence of the second Utility Card differs from the UTIL block sequence of the first card. For the correct pogo block assignment of the second card, refer to "Utility pogo block mapping" in TDC topic "Utility pogo block - Utility Card (V93000 EXA Scale systems)" (347642).

**Note** Connect all pogo pins (of all blocks) of a supply domain to a separate power plane on the DUT board. Connect all sense pins (of all blocks) of a supply domain to the power plane.

Maximum offset of utility ground to system ground: ±0.5 V

### ⚠ CAUTION Incorrect galvanic connection

Fault currents can damage electronic equipment.

➔ Do not connect utility power supply pogo pins of utility blocks together, if these utility blocks have been assigned to different internal utility modules or utility cards.

## External utility power supply

The optional external utility power supply for EXA Scale EX systems provides extra power for active components like relays, operational amplifiers, and buffer on the DUT board or probe card.

The following figure shows the external utility power supply unit.



The external utility power supply provides three power lines and remote sensing on all three lines:

- +5.0 V at 6.5 A
- +3.3 V at 4.5 A
- -5.0 V at 4.5 A

The supply is mounted inside the EXA Scale EX engineering cart.

**External utility power supply (E7002EUSR/E7002EUSM - V93000 systems)**

The external utility power supply is optional.

| Characteristic                                              |                                                                                                |                                                                                                                                                                                                                                                                                        |
|-------------------------------------------------------------|------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Output voltage                                              | 5.0 V, sensed, -0 %, +3 % (+5.15 V)                                                            |                                                                                                                                                                                                                                                                                        |
|                                                             | 3.3 V, sensed, -0 %, +3 % (+3.4 V)                                                             |                                                                                                                                                                                                                                                                                        |
|                                                             | -5.0 V, sensed, -0 %, +3 % (-5.15 V)                                                           |                                                                                                                                                                                                                                                                                        |
| Output current<br>(max. 2 A per block and for each voltage) | 5.0 V:                                                                                         | 6.5 A                                                                                                                                                                                                                                                                                  |
|                                                             | 3.3 V:                                                                                         | 4.5 A                                                                                                                                                                                                                                                                                  |
|                                                             | -5.0 V:                                                                                        | 4.5 A                                                                                                                                                                                                                                                                                  |
| Pogo pins                                                   | The optional external utility power supply is available only on the 4-row utility pogo blocks. |                                                                                                                                                                                                                                                                                        |
|                                                             | 5.0 V:                                                                                         | XPP5V and XPP5Vs                                                                                                                                                                                                                                                                       |
|                                                             | 3.3 V:                                                                                         | XPP3V3 and XPP3V3s                                                                                                                                                                                                                                                                     |
|                                                             | -5.0 V:                                                                                        | XPM5V and XPM5Vs                                                                                                                                                                                                                                                                       |
|                                                             | GND:                                                                                           | UGND and UGNDs                                                                                                                                                                                                                                                                         |
|                                                             |                                                                                                | Use only one sense line per voltage and only one ground sense line for power sensing. Leave the sense lines on the other utility blocks unconnected. If the power line is not used, leave all sense lines unconnected. This restriction applies to XPP5Vs, XPP3V3s, XPM5Vs, and UGNDs. |
| Power OK output                                             |                                                                                                | XDCGD                                                                                                                                                                                                                                                                                  |



## Miscellaneous characteristics

Side panel port

| Characteristics      | EXA Scale EX test head |
|----------------------|------------------------|
| Number of card cages | 1                      |
| DC rails             | 1                      |

Reference 10MHz

| Frequency | 10 MHz                         |
|-----------|--------------------------------|
| Impedance | 50 $\Omega$                    |
| Level     | 6 dBm (1.26Vpp) <sup>[5]</sup> |
|           | 1.26 Vpp                       |
|           | AC-coupled                     |
| Accuracy  | 1 ppm <sup>[6]</sup>           |

### DUT interface types

All test heads support the enhanced DUT Scale interface and feature the "infinite plane" concept, which means any DUT board size fits on any DUT interface.

<sup>[5]</sup> Into 50  $\Omega$

<sup>[6]</sup> Without external adjustment

## Environmental

|                                     |                                          |
|-------------------------------------|------------------------------------------|
| Operating                           | 15 °C to 30 °C (59 °F to 86 °F)          |
| Maximum rate of change              | ± 2 #/h (3.6 #/h)                        |
| Specification warranted temperature | 20 °C to 30 °C (68 °F to 86 °F)          |
| Maximum humidity at 30°C            | < 70 % relative humidity, non-condensing |
| Storage (without water)             | -40 °C to +70 °C                         |

## Power requirements

Maximum power consumption

| Test head type       | EXA Scale EX test head |
|----------------------|------------------------|
| Number of card cages | 1                      |
| Number of slots      | 3                      |
| Maximum power        | 6 kW                   |

## Cooling requirements

A cooling unit E2760FAE, E2760FAT, E2760FAX or E2760FAXV is required to cool the EXA Scale EX test head. Up to 3/6/9 EXA Scale EX test heads can be shared with one E2760FAE/E2760FAT/E2760FAX(E2760FAXV) cooling system.

### Cooling unit E2760FAE liquid-liquid cooling unit

|                                         |                                                     |
|-----------------------------------------|-----------------------------------------------------|
| Cooling capability                      | 18 kW @ 30 °C regulated secondary water temperature |
| Cooling medium                          | De-ionized water                                    |
| Primary chilled water temperature range | 5 °C to 20 °C (41 °F to 68 °F)                      |

### Cooling unit E2760FAT liquid-liquid cooling unit

|                                         |                                                    |
|-----------------------------------------|----------------------------------------------------|
| Cooling capability                      | 36 kW @ 30°C regulated secondary water temperature |
| Cooling medium                          | De-ionized water                                   |
| Primary chilled water temperature range | 5 °C to 20 °C (41 °F to 68 °F)                     |

### Cooling unit E2760FAX and E2760FAXV Liquid-liquid cooling unit

|                                         |                                                    |
|-----------------------------------------|----------------------------------------------------|
| Cooling capability                      | 54 kW @ 30°C regulated secondary water temperature |
| Cooling medium                          | De-ionized water                                   |
| Primary chilled water temperature range | 5 °C to 20 °C (41 °F to 68 °F)                     |

## Weights and dimensions

Test head weight with engineering cart and power box (fully loaded, the cooling unit is not included)

|                          |                     |
|--------------------------|---------------------|
| EXA Scale EX test system | < 120 kg [264.5 lb] |
|--------------------------|---------------------|

Dimensions of the EXA Scale EX test cell (test head installed to the engineering cart) (height x width x depth, incl. docking)

|                        |                                                        |
|------------------------|--------------------------------------------------------|
| EXA Scale EX test cell | 0.99 m x 0.73 m x 0.96 m [3.25 ft x 2.39 ft x 3.15 ft] |
|------------------------|--------------------------------------------------------|

Test head dimensions (height x width x depth)

|                        |                                                        |
|------------------------|--------------------------------------------------------|
| EXA Scale EX test head | 0.38 m x 0.19 m x 0.55 m [1.25 ft x 0.62 ft x 1.80 ft] |
|------------------------|--------------------------------------------------------|

Cooling unit E2760FAE

|                                     |                                                           |
|-------------------------------------|-----------------------------------------------------------|
| Dimensions (height x width x depth) | 0.832 m x 0.363 m x 0.815 m [2.73 ft x 1.19 ft x 2.67 ft] |
|-------------------------------------|-----------------------------------------------------------|

|            |                  |
|------------|------------------|
| Net weight | 81 kg [178.5 lb] |
|------------|------------------|

Cooling unit E2760FAT

|                                     |                                                           |
|-------------------------------------|-----------------------------------------------------------|
| Dimensions (height x width x depth) | 0.852 m x 0.443 m x 0.815 m [2.79 ft x 1.45 ft x 2.67 ft] |
|-------------------------------------|-----------------------------------------------------------|

|            |                  |
|------------|------------------|
| Net weight | 92 kg [202.8 lb] |
|------------|------------------|

Cooling unit E2760FAX

|                                     |                                                           |
|-------------------------------------|-----------------------------------------------------------|
| Dimensions (height x width x depth) | 1.095 m x 0.504 m x 1.078 m [3.59 ft x 1.65 ft x 3.54 ft] |
|-------------------------------------|-----------------------------------------------------------|

|            |                   |
|------------|-------------------|
| Net weight | 173 kg [381.4 lb] |
|------------|-------------------|

Cooling unit E2760FAXV

|                                     |                                                           |
|-------------------------------------|-----------------------------------------------------------|
| Dimensions (height x width x depth) | 1.095 m x 0.504 m x 1.078 m [3.59 ft x 1.65 ft x 3.54 ft] |
|-------------------------------------|-----------------------------------------------------------|

|            |                   |
|------------|-------------------|
| Net weight | 180 kg [396.8 lb] |
|------------|-------------------|